Group and solo flight: comparisons

What differs between group time and solo flight, and what remains when the controls are in place?

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Abstract

Group time differs from solo flight, but so do the pilots, the places and the routes of groups. This report compares paragliders in three populations, solo flights, group time and the same flights alone, and gives for each result the number of flights or pieces behind it. Raw, group time has a straighter progress over a quarter of an hour than solo flights, $H_{\mathrm{loc}}(1000\ \mathrm{s})$ of 0.893 against 0.837, and glides faster and flatter. Against the same route flown on another day by isolated solo flights of the same wing level, the gap $\Delta H_{\mathrm{loc}}(1000\ \mathrm{s})$ falls from $+0.058$ to $+0.017$ [$+0.011$, $+0.025$] on 15 539 pieces: the route explains 70% of it, for experienced pilots and beginners alike. What remains is small and goes with flying together; whether the group causes it stays open.

1 Aim

Does a paraglider move differently in a group than alone? This report answers with a series of comparisons of group time with solo flight. Each has its own control, and a result is read together with the number of flights or pieces behind it, given in every table and figure.

Section 2 defines the populations and counts them, Section 3 compares them as they are, and Section 4 compares group time with the same route flown alone.

2 Populations and measures

Which flights, which stretches of flight, and how many of them?

Groups. The labels are those of the note *Group flights in free flight* (`methods/group-identification`). Two pilots are in contact when one is on the path of the other within $\delta = 2$ min, $\varepsilon_h = 500$ m horizontally and $\varepsilon_z = 300$ m vertically (headings compared only in glide) for at least $\tau^* = 38.1$ min. The configurations of an instant are the connected components of the contact graph, and a group keeps the same members over a maximal interval. A contact means shared air within sight; it does not prove interaction.

States. After its *departure*, the first fix 3 km from the takeoff, every instant of a flight is in a group or solo; before it the pilot counts as neither. Flights that never depart are left out. We use paraglider flights logged every 10 s or finer, so that the labels exist at every instant; the phases (climb, search, glide) exist on the flights logged every second.

Populations. Three, all of paragliders.

- S, *solo flights*: flights that never entered a group, the whole of their time after the departure. They may have had contacts shorter than τ^*.
- G, *group time*: the stretches in a group of the flights that entered one at least once. A stretch stays in one group; G counts a flight once however many groups it joins.
- A, *the same flights alone*: the solo stretches of those same flights, a control with the same pilots, gliders, loggers and days as G.

The control of Section 4 is stricter than S and is defined there.

Measure. A *piece* is a stretch within one segment of the track, of at least 1250 s, on a 10-s grid of its clock. For a flight f and a population, the time-averaged mean squared displacement pools the origins of all its pieces in that population,

$$\overline{\delta^2_f}(\tau) = \frac{1}{N_f(\tau)} \sum_t \|\mathbf{r}(t+\tau) - \mathbf{r}(t)\|^2, \tag{1}$$

with t and $t+\tau$ in the same piece. The ensemble MSD(τ) is its mean over the flights, one weight per flight, and H is half the slope of log MSD against log τ over the lags from 10 to 1000 s: $H = 1$ for a straight line, $H = \frac{1}{2}$ for a random walk. $H_{\text{loc}}(\tau)$ is the same slope over the lags within ± 0.25 decades of τ. It is read at $\tau = 1000$ s, over one to two climb–glide cycles. Climbs and glides are summarised by the share of time in climb, the share of searches that end in a climb, the climb rate, the glide speed and the glide ratio, each a ratio of sums over the phase runs of a population. Bands are 5–95% of bootstrap draws that resample the pilots and the dependency blocks of the grouping.

	solo flights S	group time G	same flights alone A
flights, H and H_{loc}	119 129	33 435	29 611
pieces of at least 1250 s	125 997	45 476	47 577
flights, time in climb and searches	80 278	24 099	23 469
flights, climb rate and glides	115 629	33 406	30 703

Table 1: Members of the populations, paragliders. The flights of G and A are the same flights, and A keeps those with a solo stretch of at least 1250 s.

Sizes. Table 1 counts the members. G has 33 435 flights in 45 476 pieces and S has 119 129 flights in 125 997. A has 29 611 flights in 47 577 pieces, fewer flights than G: the others have no solo stretch long enough. The phase statistics rest on fewer flights, those logged every second.

3 The raw difference

How do the three populations differ before any control?

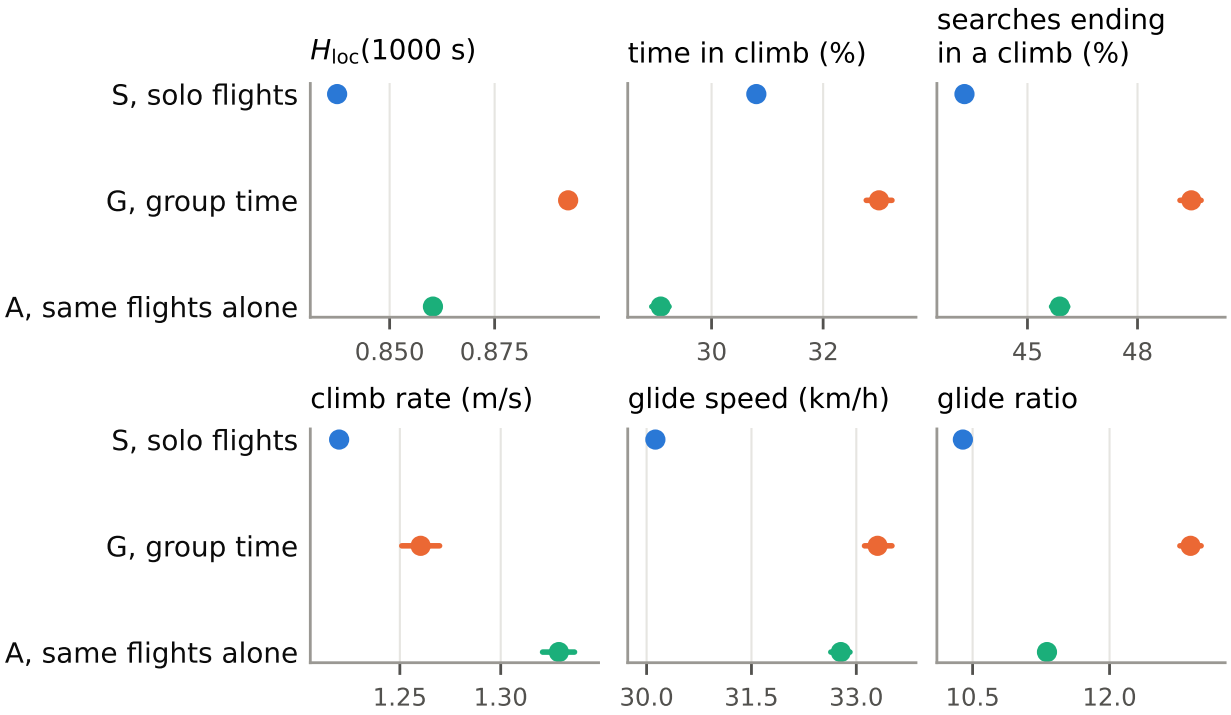


Figure 1: Pooled statistics of S, G and A with 5–95% bands. Exponents on 119 129, 33 435 and 29 611 flights, phase statistics on the flights of Table 1. Values in Table 2.

Group time progresses straighter over long lags than solo flights: $H_{\text{loc}}(1000\text{ s})$ is 0.893 in G, 0.860 in A and 0.837 in S (Figure 1). It also spends more time climbing, ends more searches in a climb and glides faster

and flatter, with a glide ratio of 12.9 against 10.4. The same flights alone sit between the two at long lags and have the highest climb rate. Part of the raw difference may belong to who flies in groups, and to where and when they fly.

	solo flights S	group time G	same flights alone A
flights (TA-MSD)	119 129	33 435	29 611
H (10–1000 s)	0.899 [0.898, 0.899]	0.916 [0.915, 0.917]	0.914 [0.913, 0.915]
$H_{\mathrm{loc}}(1000\ \mathrm{s})$	0.837 [0.837, 0.838]	0.893 [0.891, 0.894]	0.860 [0.859, 0.862]
time in climb (%)	30.8 [30.7, 30.9]	33.0 [32.8, 33.2]	29.1 [28.9, 29.2]
searches ending in a climb (%)	43.3 [43.2, 43.4]	49.5 [49.2, 49.7]	45.9 [45.7, 46.1]
climb rate (m/s)	1.22 [1.22, 1.22]	1.26 [1.25, 1.27]	1.33 [1.32, 1.34]
glide speed (km/h)	30.1 [30.1, 30.2]	33.3 [33.1, 33.5]	32.8 [32.6, 32.9]
glide ratio	10.4 [10.4, 10.4]	12.9 [12.8, 13.0]	11.3 [11.2, 11.4]
cross-country speed (km/h)	17.2 [17.2, 17.3]	20.0 [19.8, 20.3]	20.0 [19.8, 20.1]

Table 2: Pooled statistics of the populations of Table 1, with 5–95% bands. The first row is the number of flights of the exponents; the other rows rest on the flights with phases (Table 1).

What we keep. A difference to explain, largest at long lags and in the glides: group time against solo flights, with the same flights alone in between. The next section removes the route.

4 The same route flown alone

Groups may form on a ridge or a convergence line where any pilot progresses straight, and the straightness would then belong to the route and not to the group. Does the gap remain when group time and its control fly the same route?

The control is the same route flown alone, on another day. It differs from the route test of the general report (`vs-solo/general`) on three points: the control may come from any site and date other than that of the piece, and every piece of group time that has a control is used, not a sample; the control must also have had nobody within 10 km at the same time, since a flight with no contact in the grouping can still have flown near others; and a piece is compared with controls of the same wing level, with the results given for experienced and beginner pilots apart.

4.1 The control and the measure

Which flights are the same route flown alone, and how is the gap with group time measured?

Matching. The pieces of group time are those of Section 2. A segment k of a solo flight is a control of the piece g when all hold.

(i) **Route.** At least 80% of the points of g, one every 30 s, lie within 500 m of the track of the flight of k, which flew them in the same direction; k is the part of the track between its points nearest to the first and to the last of them, and lasts at least 1250 s.

(ii) **Date.** Other than that of g, at any site.

(iii) **Season and hour.** The middles of g and k are in months within one (circular) and in local hours within two.

(iv) **Wing level.** The same: beginners (A, B) or experienced (C, D, CCC) in the FFVL catalogue.

(v) **Solo.** The flight never had a contact of any length, which is stricter than S: a contact shorter than τ^* already excludes it.

(vi) **Isolation.** At every point of k, one every 30 s, no other flight of the archive, alone or in a group, paraglider or hang glider, was within 10 km within one minute (the index of the archive holds one position per minute).

All the matches are kept. 19 786 pieces have a match under the first five conditions and 15 539 after the isolation; they have 85 524 matches, a median of 3 per piece, from 5 648 solo flights of 2 000 pilots, against 3 189 pilots on the group side.

Comparison. In analogy with the zones of the general report, the stratum is the piece. The control of a piece is the mean of the MSD of its matches, and each piece weighs the same on both sides. With n pieces

and m_g the matches of the piece g,

$$\mathrm{MSD}_{\mathrm{G}}(\tau) = \frac{1}{n}\sum_{g}\mathrm{MSD}_g(\tau), \qquad \mathrm{MSD}_{\mathrm{ctrl}}(\tau) = \frac{1}{n}\sum_{g}\frac{1}{m_g}\sum_{k\in g}\mathrm{MSD}_k(\tau), \tag{2}$$

and both sides are means of many MSDs before their logarithm. The gap is $\Delta H_{\mathrm{loc}}(\tau) = H^{\mathrm{G}}_{\mathrm{loc}}(\tau) - H^{\mathrm{ctrl}}_{\mathrm{loc}}(\tau)$, positive when group time progresses straighter. The baseline compares the same pieces with the solo flights of their site-day and hour, so that the two gaps refer to the same group time; the share of the baseline gap that the route explains is

$$1 - \frac{\Delta H^{\mathrm{route}}_{\mathrm{loc}}(1000\ \mathrm{s})}{\Delta H^{\mathrm{base}}_{\mathrm{loc}}(1000\ \mathrm{s})}. \tag{3}$$

Uncertainty. Bands are 5–95% of 1000 bootstrap draws that resample the pilots and the dependency blocks of the grouping, independently, and recompute every statistic; each difference has its own band. The thin lines of the figures are the range when one year at a time is left out. The baseline and the explained share are point estimates.

What we keep. The control is the same route, on another day, in the same season, hour and wing level, flown by a solo pilot with nobody within 10 km. One gap, ΔH_{loc}, is read at 1000 s and compared with the baseline of the same pieces.

4.2 What remains of the gap

How much of the straighter progress of groups is left once the route is the same?

Over a quarter of an hour, group time still progresses straighter than the same route flown alone: $\Delta H_{\mathrm{loc}}(1000\ \mathrm{s})$ is $+0.017$ [$+0.011$, $+0.025$], against $+0.058$ for the same pieces against the solo flights of their site-day and hour (Table 3). Fixing the route lowers the gap by 70%, Eq. (3); its band stays away from zero. The long series, on the pieces that last long enough, gives about the same gap up to 3000 s, with bands wide enough to reach zero (Figure 2).

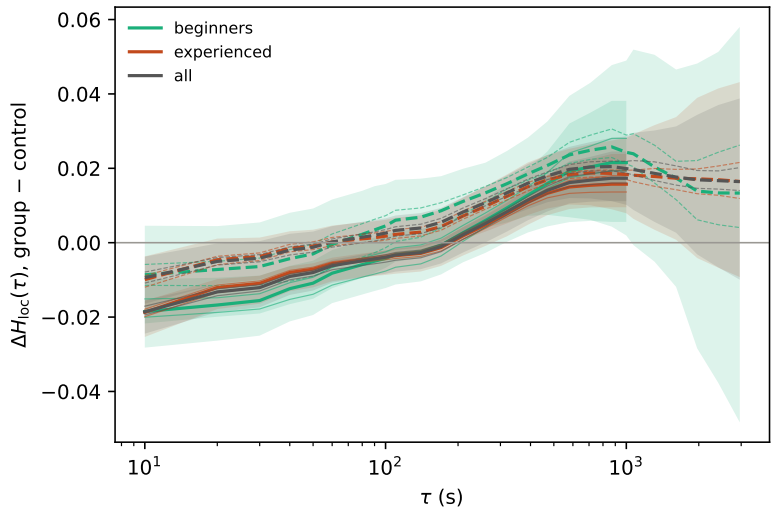


Figure 2: $\Delta H_{\mathrm{loc}}(\tau)$, group time minus the same route flown alone, for beginners (4 715 pieces), experienced pilots (10 824) and both (15 539). Continuous: the lags up to 1000 s; dashed: the long series, on the pieces that last long enough. Bands 5–95%, thin lines the range when one year is left out.

The gap changes sign with the lag: at short lags, within one climb or glide, group time is less straight than the same route flown alone, and it is more straight from about 200 s on (Figure 2).

What we keep. The route, with the season, the hour and the wing level, explains about 70% of the long-lag gap between group and solo flight. The rest, $+0.017$ at 1000 s, is small and goes with flying together.

4.3 Experienced pilots and beginners

Does the route explain the same share for both wing levels?

wing level	pieces	matches	against site-day and hour	against isolated, same route	explained
all	15 539	85 524	+0.058	+0.017 [+0.011, +0.025]	70%
experienced	10 824	64 242	+0.051	+0.016 [+0.010, +0.024]	69%
beginners	4 715	21 282	+0.068	+0.022 [+0.006, +0.038]	68%

Table 3: $\Delta H_{\mathrm{loc}}(1000\ \mathrm{s})$ of group time against the solo flights of the same site-day and hour, and against the isolated solo flights on the same route, with the 5–95% band; the share of the first that the route explains, Eq. (3).

It does (Table 3, Figure 3). For the 10 824 pieces of experienced pilots $\Delta H_{\mathrm{loc}}(1000\ \mathrm{s})$ falls from +0.051 to +0.016 [+0.010, +0.024], 69% explained. For the 4 715 pieces of beginners it falls from +0.068 to +0.022 [+0.006, +0.038], 68% explained. The beginners' baseline gap is larger and their band wider, since they have fewer pieces; the two levels differ by one point in the share explained, well within their bands.

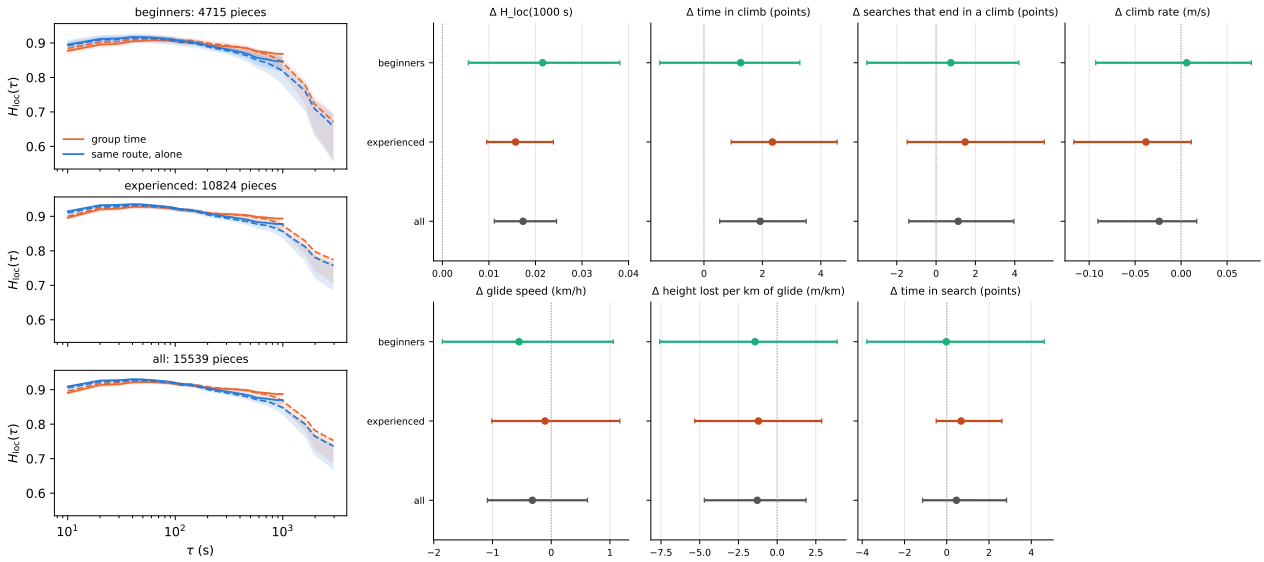


Figure 3: Left: $H_{\mathrm{loc}}(\tau)$ of group time (orange) and of the same route flown alone (blue), by wing level, with the number of pieces in the panel titles; dashed, the long series. Right: the difference group minus control of $H_{\mathrm{loc}}(1000\ \mathrm{s})$ and of the statistics of climbs and glides, with 5–95% bands, by wing level.

What we keep. Experienced pilots and beginners give the same share explained, so that the route accounts for the same part of the gap whatever the wing level.

4.4 Climbs and glides

Does a group climb and glide differently from the same route flown alone?

Table 4 compares the time in each phase and the statistics of climbs and glides of the pieces that have phases (loggers of 1 s) with the same statistics on their controls. Group time has more time in climb and less in glide, and its pilots start more climbs per hour; the climb rate, the glide speed, the glide ratio and the height lost per kilometre of glide differ by less than their bands. In a group the pilot spends more time climbing, and the climb itself is not faster.

The right of Figure 3 gives the same differences for each level: the bands of the beginners are wider and contain zero for every statistic of the phases.

What we keep. On the same route a group spends more time climbing and less gliding; the speed, the rate and the glide ratio do not differ.

5 Conclusions and limits

What can be concluded, and what cannot?

What we keep.

all levels	group time	same route alone	difference
pieces, segments	10 736	42 685	
time in glide (%)	57.5	59.9	-2.4 [-5.0, -1.0]
time in search (%)	13.7	13.2	$+0.5$ [-1.1, $+2.8$]
time in climb (%)	28.8	26.9	$+1.9$ [$+0.5$, $+3.5$]
searches ending in a climb (%)	49.6	48.5	$+1.1$ [-1.4, $+4.0$]
climbs per hour	7.46	6.99	$+0.48$ [$+0.20$, $+0.94$]
climb rate (m/s)	1.12	1.14	-0.02 [-0.09, $+0.02$]
glide speed (km/h)	28.6	28.9	-0.3 [-1.1, $+0.6$]
glide ratio	16.2	15.9	$+0.3$ [-0.5, $+1.2$]
height lost per km of glide (m/km)	61.6	62.8	-1.3 [-4.7, $+1.9$]

Table 4: Climbs and glides of group time and of the same route flown alone, both wing levels together, on 10 736 pieces of group time with phases and 42 685 matched segments; the difference group minus control with its 5–95% band. Times are shares of the time flown, in %, and their differences are in percentage points.

- Raw, group time (G, 33 435 flights) progresses straighter than solo flights (S, 119 129) over long lags, and the same flights alone (A, 29 611) sit in between.
- The route, with the season, the hour and the wing level, explains 70% of the gap $\Delta H_{\mathrm{loc}}(1000\ \mathrm{s})$ between group time and solo flight; the same for experienced pilots and beginners.
- What remains, $+0.017$ [$+0.011$, $+0.025$], is small, with a band away from zero; at short lags group time is less straight than the control.
- On the same route a group spends more time climbing and less gliding; the climb rate and the glide speed do not differ.

Limits.

- The raw comparison fixes nothing: S and G differ in pilots, places, days and hours.
- The days differ: season and hour are matched, the weather of the day is not.
- The controls flew with nobody within 10 km. They may differ from the pilots who fly in groups, and this test has no selection contrast to measure it.
- Only the 15 539 pieces whose route was flown alone on another day enter; their routes are frequented ones.
- The bands measure the sampling of pilots and blocks. The radius and the cover of the route were varied only in the general report, on a sample, and the isolation distance not at all.

Fixing the route removes most of the straighter progress of groups; whether what remains is caused by the group stays open.